

Performance Testing & Model Evaluation

Implement Client Script & UI Policy (Incident): Functional, Performance, Model Metrics & Documentation Summary

Date	3 October 2026	Team ID	SWTID-2026-3878
Project Name	Implement Client Script & UI Policy (Incident)	Platform	ServiceNow (Incident table)
Team Leader	Ashwini V	Team Size	5 members
Phase	Project Development Phase	Marks	Performance Test: 10 Marks

1. Executive Summary & Project Environment

This report evaluates the functional correctness, response time and consistency of the ServiceNow Incident form after the implementation of a UI Policy and Client Scripts (onLoad, onChange, onSubmit, onCellEdit). It also records model-evaluation details (not applicable to this rule-based project), the fine-tuning carried out after testing, and a summary of the project documentation. The team comprises Ashwini V (Team Leader), Gunasri E, Dhivyabharathi G, Asanciya V and Niveatha J.

Component	Scope of Evaluation
UI Policy	Conditions and actions on the Incident table that make fields mandatory, read-only or visible, with Reverse if false, Global and On load enabled.
Client Scripts	onLoad notice for P1 incidents, onChange guidance, onSubmit validation (Resolution notes of at least 10 characters) and onCellEdit blocking on list views, using g_form and g_user.
Platform & Database	ServiceNow application server and managed database; the existing Incident table (extends Task) is used with no new tables or custom backend code.
Security & Access	Username/password sign-in with roles (itil, admin); server-side ACLs remain the real protection, while UI Policies and Client Scripts guide users on the form.
Model Performance	Not applicable (rule-based project, no machine learning model). Equivalent test metrics are reported in Section 4.

2. Test Scenarios & Service-Level Targets

Scenario ID	Test Area	Test Method	Expected Behaviour / SLA
FT-01	Client Script field validation	Enter valid and invalid values on a new Incident	Error shown for invalid input; valid input accepted
FT-02	UI Policy: mandatory fields	Set State = Resolved, save with and without fields	Save blocked until Close code and notes are filled
FT-03	UI Policy: read-only / visible	Change State, Category and Impact to make condition true, then false	Fields lock or show, and revert when false
FT-04	Trigger conditions and Reverse if false	Test true, false and edge values	Applies and reverses with no script conflict
PT-01	Form load time	Open the form 5 times; time with browser developer tools	Load time $\leq$ 3.0 seconds
PT-02	Response on field change	Change multiple fields rapidly and measure	Update time $\leq$ 1.0 second
PT-03	Multi-user / multi-record	5 users update Incident records at the same time	100% consistent, no slowdown

3. Empirical Test Results

Test ID	Area	Measured Result	Target	Status
FT-01	Client Script validation	Invalid input rejected; valid accepted	Correct validation	PASSED
FT-02	Mandatory fields	Save blocked, then succeeded when filled	Correct enforcement	PASSED
FT-03	Read-only / visible	Applied and reverted correctly	Correct reversal	PASSED
FT-04	Trigger conditions	No conflicts on true / false / edge	No conflict	PASSED
PT-01	Form load time	1.84 s (average of 5 runs)	≤ 3.0 s	PASSED
PT-02	Field-change response	0.32 s (average)	≤ 1.0 s	PASSED
PT-03	Multi-user (5 users, 25 updates)	25 of 25 consistent; +0.12 s peak delay	100% consistent	PASSED

Performance Margins vs Target

Test ID	Target	Measured	Margin vs Target
PT-01	≤ 3.0 sec	1.84 sec	1.16 sec (38.7%) faster
PT-02	≤ 1.0 sec	0.32 sec	0.68 sec (68.0%) faster
PT-03	100% consistent	100% (25/25)	Target met, 0 conflicts

All seven test scenarios met their targets. Values are recorded from test runs on the development instance.

4. Model Performance & Calculations

Regression / Classification Model: not applicable, as the project uses a rule-based UI Policy and Client Scripts. Equivalent metrics are the five test activities (Mandatory Enforcement, Successful Save, Reverse Condition, List Edit Blocking and Form-Based Update): **5 total, 5 passed, 0 failed, pass rate 100%**. Validation method: functional testing of each activity on the form and list view, followed by User Acceptance Testing with positive and negative cases.

1. Test Pass Rate (PR)

Share of executed test scenarios that passed.

$$\text{PR} = \frac{\text{Passed}}{\text{Total}} \times 100\% = \frac{7}{7} \times 100\% = \mathbf{100\%}$$

2. Average Form Load Time ($\bar{t}$)

Mean of five timed loads: 1.92, 1.78, 1.81, 1.85 and 1.84 seconds.

$$\bar{t} = \frac{\sum t_i}{n} = \frac{9.20 \text{ s}}{5} = \mathbf{1.84 \text{ seconds}}$$

3. Fine-Tuning Efficiency Gain (η)

Improvement in form load time after script and condition fixes (Section 5).

$$\eta = \frac{t_{\text{before}} - t_{\text{after}}}{t_{\text{before}}} \times 100\% = \frac{2.60 \text{ s} - 1.84 \text{ s}}{2.60 \text{ s}} \times 100\% = \mathbf{29.2\%}$$

5. Fine-Tuning & Issue Diagnostics

Hyperparameter tuning is not applicable. Here, tuning means adjusting conditions and script messages after test findings.

Area	Issue Identified	Fix Implemented
UI Policy order	Two policies on State competed for the same field, giving unpredictable results.	Set clear Order values so each field has one winning policy.
onChange script	Repeated g_form calls caused a short delay on rapid field changes.	Cached field values and removed redundant calls.
onSubmit validation	Short Resolution notes could be saved from the list view.	Added onCellEdit blocking and the 10-character check on submit.
Reverse if false	Read-only state stayed after changing State back from Closed.	Enabled Reverse if false and re-tested the condition.
Instance values	State choice values may differ between instances.	Verified values after moving changes through an Update Set.

6. Project Documentation Summary

Item	Details
Setup	Free Personal Developer Instance; admin user; create the UI Policy (All > System UI > UI Policies) and Client Scripts (All > System Definition > Client Scripts).
Configuration records	sys_ui_policy, sys_ui_policy_action and sys_script_client; changes captured in an Update Set.
Client APIs used	g_form.getValue, setMandatory, setReadOnly, setDisplay, showFieldMsg and addInfoMessage.
Known limits	Rules run in the browser only. Imports, web services and server scripts are not checked.
Future work	Business Rules or Data Policies, extension to Problem and Change tables, Catalog Client Scripts.

7. User Acceptance Testing (UAT) Sign-Off Matrix

UAT Module	Acceptance Criteria	Reviewer Feedback	Sign-Off
UI Policy	Fields turn mandatory, read-only or visible and revert correctly.	Behaviour verified on the Incident form.	PASSED / SIGNED
Client Scripts	onChange and onSubmit show clear messages and block invalid saves.	Messages clear and helpful.	PASSED / SIGNED
List and Form Updates	List edits are blocked; form updates save valid data.	Verified on both views.	PASSED / SIGNED
Performance	Load time ≤ 3 s; field response ≤ 1 s.	Within limits in all runs.	PASSED / SIGNED

Final Conclusion & Sign-Off Recommendation

The functional, performance and acceptance tests confirm that the UI Policy and Client Scripts on the Incident form meet all targets, with a 100% pass rate across seven scenarios, a 1.84 s average load time and a 29.2% efficiency gain after fine-tuning. The project is recommended for sign-off.